

Lab: Case Study Analysis — Precision and Its Disorders

Objective

Analyze clinical cases involving aberrant precision weighting, applying the Active Inference framework to interpret attentional and cognitive disorders.

Part 1: ADHD and Sustained Attention

Case: Patient T.J., 12, diagnosed with ADHD-Combined type. In the classroom, T.J. shifts attention every few minutes — from the teacher to the window to a classmate's pencil to his own shoelaces. He performs well on novel, stimulating tasks but cannot sustain focus on routine work. Methylphenidate (Ritalin) improves his sustained attention significantly.

- Interpret ADHD in Active Inference terms: what precision imbalance explains the difficulty sustaining attention on routine tasks?
- Why does novelty *help* T.J. focus? (Hint: novel stimuli generate high prediction error, which temporarily captures precision allocation.)
- How does methylphenidate (a dopamine/norepinephrine reuptake inhibitor) address the precision deficit?

Write your response here

Part 2: Autism and Sensory Precision

Case: Patient A.R., 22, diagnosed with autism spectrum condition. She reports extreme sensitivity to fluorescent lighting (which she describes as "constant flickering that others don't

notice"), difficulty filtering background noise in conversations, and a remarkable ability to detect subtle visual patterns (she works as a quality inspector in a manufacturing plant).

- Apply Lawson et al.'s (2014) high-precision hypothesis: how does elevated sensory precision explain both A.R.'s sensory sensitivities and her pattern-detection strengths?
- Why might high sensory precision lead to difficulty integrating contextual information (top-down priors)?
- How does this framework avoid pathologizing autism by recognizing that high precision is a different cognitive profile, not simply a deficit?

Write your response here

Part 3: Generalized Anxiety Disorder

Case: Patient M.N., 35, with generalized anxiety disorder. She reports constant worry about health, finances, and relationships. She describes an inability to "turn off" the scanning for potential threats. Physical symptoms include muscle tension, rapid heartbeat, and difficulty sleeping.

- Interpret chronic anxiety as excessive precision on threat-related prediction errors. What does this mean for M.N.'s generative model?
- How do the physical symptoms (muscle tension, rapid heartbeat) relate to Active Inference? (Hint: the body is acting to "confirm" the threat prediction through autonomic activation.)
- What therapeutic approach does this framework suggest? (Consider interventions that down-regulate precision on threat prediction errors — e.g., cognitive restructuring, interoceptive exposure.)

Write your response here

Part 4: Pharmacological Precision Modulation

Case: A double-blind study administers donepezil (acetylcholinesterase inhibitor, increasing ACh) to healthy participants. Results show improved performance on a visual discrimination task (detecting faint gratings embedded in noise) but impaired performance on a contextual integration task (identifying the Kanizsa triangle from partial contours).

- Interpret these results in Active Inference terms: how does increased ACh (bottom-up precision) improve discrimination but impair context-dependent perception?
- What does this tell us about the trade-off between sensory precision and prior influence?
- Predict the results you would expect with a dopamine agonist instead of a cholinergic agonist.

Write your response here

Part 5: Synthesis

Write a 200-word synthesis explaining how these four cases collectively demonstrate that cognition is mediated by precision — and that many cognitive and psychiatric conditions can be understood as specific patterns of precision dysregulation.

Write your response here

Lab Summary

Part	Skill Developed	Case Type
1	Clinical reasoning	ADHD and dopaminergic precision
2	Strength-based analysis	Autism and sensory precision
3	Therapeutic reasoning	Anxiety and threat precision
4	Pharmacological analysis	Drug effects on precision dynamics
5	Integrative synthesis	Connecting precision disorders to a unified framework